Evidence-Aware Risk Scoring for Scientific Claim Verification

One-Page Research Summary

Research question. Can a risk estimator that observes both NLI behavior and label-free evidence-retrieval signals rank scientific claim-verification errors better than an NLI-only risk model?

Motivation. A verifier may be highly confident even when retrieval, sentence selection, NLI, or claim aggregation fails. Selective prediction studies whether high-risk outputs can be rejected while retaining a useful fraction of predictions. It does not make the underlying classifier more accurate.

Data and system. I used the official SciFact data and audited provenance, schema, licenses, duplicates, and leakage risks. The frozen E1 pipeline used BM25 retrieval, label-free lexical sentence selection, and a pinned DistilRoBERTa NLI cross-encoder. Train was divided by connected cited-document groups into 527 fit, 141 calibration, and 141 Internal Test claims. Calibration was not used in this study.

Method. Full EARS is an L2 logistic-regression error-ranking model. Its frozen inference-time features represent final-decision confidence, support-contradiction margin, predicted category, BM25 top-two margin, and evidence conflict. B1 is a fair NLI-only comparator trained on the same 329 eligible fit claims with the same target, preprocessing, regularization, and solver. The only method difference is the evidence features.

Independent result. The Internal Test contained 88 eligible SUPPORT/CONTRADICT claims. The base verifier achieved 0.488636 accuracy. EARS improved AURC from 0.300833 to 0.264987 relative to B1. The paired difference was -0.035846, with 95% grouped-bootstrap CI [-0.076087, -0.003447]. At the preregistered 80% coverage point, however, EARS was worse: selective risk 0.457143 versus 0.428571, a difference of +0.028571 with CI [0.000000, 0.063492]. EARS accepted 32 errors among 70 claims; B1 accepted 30.

Conclusion. H1 was **inconclusive under the preregistered joint criterion** because both endpoints had to improve. EARS changed error ranking and improved average prefix behavior, but did not reliably improve the fixed operating point. H2 evidence-degradation robustness and H3 probability calibration were not evaluated.

Integrity and limitations. Models and predictions were hashed before test-label access. Confidence intervals used 10,000 paired connected-group bootstrap replicates. Two deterministic recalculations produced byte-identical core results. The test sample was small, the base verifier made 45/88 errors, NO_RELATION was not a ground-truth insufficiency label, and the Internal Test was not an external dataset. This system is not suitable for medical or other high-stakes fact checking.

Verifiable contribution. I developed and documented the research question, staged protocol, leakage controls, frozen comparison, reproducibility checks, error analyses, and restrained interpretation. AI-assisted tools supported implementation and drafting; all public claims remain subject to my review and explanation.